On the origin of the factor u of the P18 profile: what is established, what is excluded, what remains open

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companion note to *From a primordial fluctuation to an idealised rotation curve* (pf2irc), Sects. 2 and 6.5

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Abstract

The P18 profile is the singular isothermal sphere (SIS) times u^2, with $u = 1 + (x-1)e^{-x}$ and $x = R/r_c$. Its form is empirical. This note collects what a first inquiry into its origin has achieved. *Established (exact).* u is the unique regular solution of the critically damped equation $u'' + 2u' + u = 1$, $u(0) = 0$, $u'(0) = 2$, the member $b = 1$ of the regular family $u_b = 1 + (bx-1)e^{-x}$. If $\mu(x)$ is the mass fraction of an exponential disc of scale length $R_d = r_c$ inside x, then $u = 1 - \mu''$: P18 deviates from the SIS by minus the radial gradient of the disc's mass per unit radius. The double root of the equation is the composition of the exponential surface density with the annulus factor $2\pi R$. The value $b = 1$ is the only one for which the first-order part of the density deviation carries no mass; the mass excess $c = \frac{1}{4} = \int (u-1)^2 \mathrm{d}x$ is purely second order, and the displaced mass splits exactly as $x - m = 2\mu' - \int_0^x \mu''^2$. At the temperature of the SIS, the Jeans equation requires an anisotropy $\beta = 1 - xu'/u - m/x$: isotropic at the centre, mildly radial ($\beta \simeq 0.15$) between 0.5 and $3\,r_c$, isotropic far out. *Physical setting.* In the collapse of the paper the dark matter stays nearly spherical ($v_{\rm rot}/\sigma \lesssim 4\lambda$), while the gas settles into a disc at its centrifugal radius, $R_d/r_c \simeq 0.7$–1.4. If the dwarfs kept all their baryons, this disc would dominate the centre; about 90% must be expelled. *Excluded.* Two numerical experiments test whether the expelled gas, through the changing potential, can carve P18 out of the dark matter. Bursts in the paper's self-similar shells flatten the cusp but act on the scale of the accretion front. Bursts of an exponential disc in an isotropic SIS change little within the baryon budget and, even far beyond it, expand the halo without removing the cusp: the centre of an SIS is shielded, because the enclosed disc mass relative to the dark mass vanishes as $r/2R_d$. *Open.* A core that forms while the disc grows during settling, so that the cusp never forms; and torques of a rotating, non-axisymmetric disc. Both are left to a future project.

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1 Purpose and results

In the paper the total matter of a late-type dwarf galaxy follows the P18 profile,

$$\rho(R) = \frac{v^2 u(x)^2}{4\pi G R^2}, \qquad u(x) = 1 + (x-1)\,e^{-x}, \qquad x = \frac{R}{r_c}, \tag{1}$$

with central density $\rho(0) = v^2/(\pi G r_c^2)$, enclosed mass $(v^2 r_c/G)\,m(x)$ with $m' = u^2$, and the closed form $m(x) = x(1-e^{-x})^2 + \frac{1}{4}[1-(1+2x+2x^2)e^{-2x}]$. For $u \equiv 1$ the profile is the singular isothermal sphere (SIS). The *displaced mass* $F(x) = x - m(x)$ is the mass that P18 lacks inside x compared with the SIS; $F' = 1 - u^2$. The paper takes the shape of u from observations. This note asks where it could come from, and records what has been achieved. Throughout, $D \equiv \mathrm{d}/\mathrm{d}x$, $g \equiv u - 1$, masses are in units of $v^2 r_c/G$ and specific energies in units of v^2 unless stated otherwise.

Table 1: Results of this note and their status.

Sect.	Result	Status
2	u is the regular, critically damped solution with $b = 1$	theorem
2	the isothermal sphere approaches the SIS underdamped in $\ln x$	theorem, numbers
3	$u = 1 - \mu''$ for an exponential disc with $R_d = r_c$; the double root is profile × annulus	theorem
4	$b = 1 \Leftrightarrow$ no mass at first order; $c = \int g^2 = \frac{1}{4}$; $F = 2\mu' - \int \mu''^2$	theorem
5	first order: shells displaced by $2R\,e^{-R/r_c}$, energy $2v^2 e^{-R/r_c} \propto \Sigma_d$	interpretation
6	Jeans: $\beta = 1 - xu'/u - m/x$ at $\sigma_r^2 = v^2/2$	theorem (target, not cause)
7	in the collapse: spherical dark matter, gas disc with $R_d \approx r_c$, 90% of the baryons expelled	estimates from the paper
8.1	bursts in the self-similar shells: cusp flattened, but on the scale of the front	numerical, negative
8.2	disc bursts in an isotropic SIS: no core; the centre is shielded	numerical, negative
8.3	shielding: enclosed disc mass / dark mass $\to M_d\,r/(2R_d)$	theorem

2 The differential equation of u

Proposition 1. *$u(x) = 1 + (x-1)e^{-x}$ is the unique solution on $[0,\infty)$ of*

$$u'' + 2u' + u = 1, \qquad u(0) = 0, \qquad u'(0) = 2. \tag{2}$$

Equivalently $g = u - 1$ satisfies $(D+1)^2 g = 0$, $g(0) = -1$, $g'(0) = 2$.

Proof. For $g = (x-1)e^{-x}$: $g' = (2-x)e^{-x}$, $g'' = (x-3)e^{-x}$, so $g'' + 2g' + g = (x - 3 + 4 - 2x + x - 1)e^{-x} = 0$, and $u = 1 + g$ satisfies the inhomogeneous equation; $u(0) = 0$, $u'(0) = 2$. A linear initial-value problem with constant coefficients has exactly one solution. □

The characteristic polynomial is $(s+1)^2$: a double root, i.e. a *critically damped* oscillator, with x in the role of time and r_c as damping length.

Proposition 2 (the regular family)**.** *Every solution of $u''+2u'+u=1$ is $u=1+(A+Bx)e^{-x}$. The density $\rho\propto u^2/x^2$ is finite at the centre if and only if $A=-1$. The regular solutions are therefore*

$$u_b(x)=1+(bx-1)\,e^{-x},\qquad u_b'(0)=b+1. \tag{3}$$

If the equation is written with a damping length λ for the dimensional radius, its regular solutions are $u_b(R/\lambda)$, with $\rho(0)=v^2(b+1)^2/(4\pi G\lambda^2)$; the core radius defined by $\rho(0)=v^2/(\pi G r_c^2)$ is $r_c=2\lambda/(b+1)$. P18 ($b=1$) is the only member with $\lambda=r_c$.

Proof. The general solution is the particular solution 1 plus the homogeneous solutions $(A+Bx)e^{-x}$. Since $u(0)=1+A$, a finite limit of u^2/x^2 requires $1+A=0$; then $u\simeq(B+1)x$ and $\rho(0)=v^2(B+1)^2/(4\pi G\lambda^2)$. Equating with $v^2/(\pi G r_c^2)$ gives $r_c=2\lambda/(b+1)$, which equals λ only for $b=1$. □

Proposition 3 (properties)**.** *(i) For $b>0$ the density crosses that of the SIS at $x=1/b$ only. (ii) u_b exceeds 1 somewhere iff $b>0$; it then has one maximum, $1+b\,e^{-(b+1)/b}$ at $x=(b+1)/b$. (iii) $m(x)-x\to c(b)=\int_0^\infty(u_b^2-1)\,\mathrm{d}x=b^2/4+3b/2-3/2$, positive iff $b>\sqrt{15}-3=0.873$. (iv) $u_b=(b+1)x-(b+\frac12)x^2+O(x^3)$, so the density falls linearly at the centre unless $b=-\frac12$. For P18: crossing at $x=1$; maximum $u(2)=1+e^{-2}=1.135$; $c=\frac14$; $F(1)=0.520$, $F=0$ at $x=3.29$, $F(\infty)=-\frac14$; the mass added outside r_c is $\int_1^\infty(u^2-1)\,\mathrm{d}x=0.770$.*

Proof. (i) u_b-1 has the sign of $bx-1$. (ii) $u_b'=(b+1-bx)e^{-x}$ vanishes once, at $(b+1)/b$. (iii) With $\int_0^\infty x^n e^{-x}\mathrm{d}x=n!$ and $\int_0^\infty x^n e^{-2x}\mathrm{d}x=n!/2^{n+1}$: $\int 2g=2(b-1)$, $\int g^2=b^2/4-b/2+1/2$; the sum is $c(b)$. (iv) Expand e^{-x}. The numbers are evaluated in `code/identities.py`. □

Proposition 4 (the isothermal sphere)**.** *The non-singular isothermal sphere, $\psi''+2\psi'/\xi=e^{-\psi}$, $\rho=\rho_0e^{-\psi}$, linearised about its singular solution $e^{-\psi_s}=2/\xi^2$ (the SIS), obeys $\delta_{ss}+\delta_s+2\delta=0$ in $s=\ln\xi$, with roots $-\frac12\pm\frac{\sqrt7}{2}i$. Its density approaches the SIS as $\xi^{-1/2}\cos(\frac{\sqrt7}{2}\ln\xi+\varphi)$: underdamped in $\ln R$, without a length scale.*

Proof. Insert $\psi=\psi_s+\delta$, use $e^{-\psi}\simeq(2/\xi^2)(1-\delta)$, and change variable to s; the equation has constant coefficients in s. □

For the same central density ($r_c=\sqrt8\,r_0$, $r_0^2=\sigma^2/4\pi G\rho_0$, $\sigma^2=v^2/2$) the first maximum of $\rho/\rho_{\rm SIS}$ is 1.661 at $x=1.44$, the first minimum 0.832 at $x=13.3$. Plain thermalisation therefore gives a scale-free ringing; a mechanism for P18 must carry its own length.

3 The disc representation of u

Definition 1. An exponential disc has $\Sigma(R)=\Sigma_0e^{-R/R_d}$ and mass $M_d=2\pi\Sigma_0R_d^2$. With $x=R/R_d$, its normalised enclosed mass is $\mu(x)=M_d(<R)/M_d$, and $\hat\Sigma=e^{-x}$ is its normalised surface density.

Lemma 1. $\mu(x)=1-(1+x)e^{-x}$, $\mu'=x\,e^{-x}$, $\mu''=(1-x)e^{-x}$, $\mu'''=(x-2)e^{-x}$. *Here $\mu'=(R_d/M_d)\,2\pi R\,\Sigma(R)$ is the mass of the annulus at R per unit x.*

Proof. $M_d(<R)=2\pi\Sigma_0\int_0^R R'e^{-R'/R_d}\mathrm{d}R'=2\pi\Sigma_0R_d^2[1-(1+x)e^{-x}]$. Differentiating: $\mu'=-e^{-x}+(1+x)e^{-x}=xe^{-x}$; $\mu''=e^{-x}-xe^{-x}$; $\mu'''=-e^{-x}-(1-x)e^{-x}$. Further $\mu'=(R_d/M_d)\,\mathrm{d}M_d/\mathrm{d}R$ and $\mathrm{d}M_d/\mathrm{d}R=2\pi R\Sigma$. □

Proposition 5. *If $R_d=r_c$, then*

$$u=1-\mu'', \tag{4}$$

equivalently $u=\mu+2\mu'$, and for the whole regular family $u_b-1=-\mu''+(b-1)\,\mu'$.

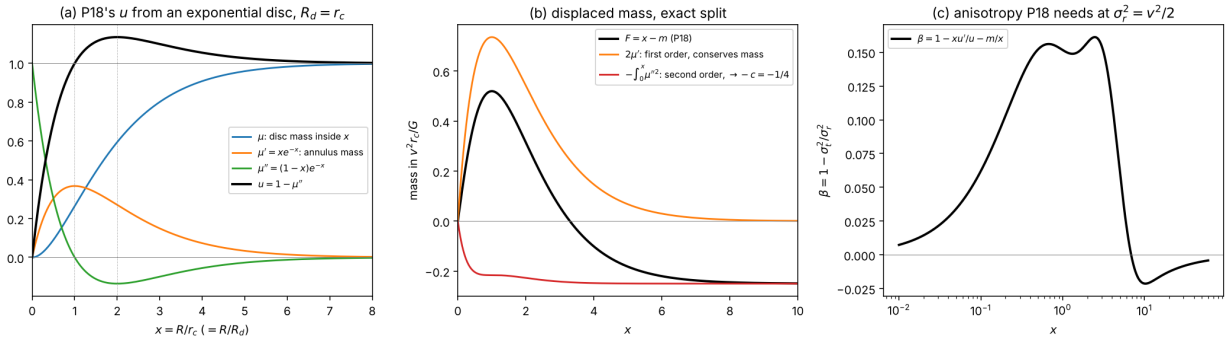


Figure 1: (a) The exponential disc with $R_d = r_c$: enclosed mass μ, annulus mass μ', its gradient μ'', and $u = 1 - \mu''$. Dotted: the crossing ($x = 1$) and the maximum ($x = 2$). (b) The exact split of the displaced mass, Proposition 8. (c) The anisotropy P18 requires at the temperature of the SIS, Proposition 9. Script: `code/fig_disc.py`.

Proof. $1 - \mu'' = 1 - (1-x)e^{-x} = 1 + (x-1)e^{-x} = u$. Since μ itself solves $\mu'' + 2\mu' + \mu = 1$ (it is the member $b = -1$ of the family), $1 - \mu'' = \mu + 2\mu'$. Finally $-\mu'' + (b-1)\mu' = (x - 1 + bx - x)e^{-x} = (bx-1)e^{-x} = u_b - 1$. □

Corollary 1 (geometric reading)**.** *With $R_d = r_c$: P18 lies below the SIS ($u < 1$) where the annulus mass μ' still grows ($x < 1$); it crosses the SIS where the annulus mass is largest ($x = 1$, $\mu'' = 0$); its overshoot ($u > 1$) lies where the annulus mass falls, and is largest at the inflection of the annulus mass ($x = 2$, $\mu''' = 0$).*

Proposition 6 (origin of the double root)**.** $(D+1)\hat{\Sigma} = 0$ *and* $(D+1)\mu' = \hat{\Sigma}$. *Hence* $(D+1)^2\mu' = 0$, *and* $(D+1)^2(u-1) = -(D+1)^2\mu'' = -D(D+1)^2\mu' = 0$.

Proof. $(D+1)e^{-x} = 0$; $(D+1)(xe^{-x}) = e^{-x} - xe^{-x} + xe^{-x} = e^{-x}$. D commutes with $(D+1)^2$. □

The two equal "relaxations in series" of the critically damped equation are thus concrete: the exponential fall of the surface density, and the annulus factor $2\pi R$, which makes the annulus mass relax towards the surface density with the same length. The factor x in xe^{-x} is the geometry of the annulus.

Remark. The solution space of Eq. (2) is the affine space $1 + \mathrm{span}\{e^{-x}, xe^{-x}\}$, so any two of its members are related by some first-order operator. The content of Proposition 5 is therefore not that a disc *causes* u, but that P18 and an exponential disc share the same damped mode with the same length, if and only if $R_d = r_c$.

4 First-order mass conservation and the second-order excess

Write $\rho/\rho_{\mathrm{SIS}} = u^2 = 1 + 2g + g^2$: a first-order and a second-order deviation.

Proposition 7. *For the regular family, $\int_0^\infty g_b \,\mathrm{d}x = b - 1$ and $\int_0^\infty g_b^2 \,\mathrm{d}x = b^2/4 - b/2 + 1/2$, so that $c(b) = 2\int g_b + \int g_b^2$. Hence $b = 1$ is the only member whose first-order deviation carries no mass, and for P18 the mass excess is purely second order:*

$$c = \int_0^\infty (u-1)^2 \,\mathrm{d}x = \tfrac{1}{4}. \tag{5}$$

Proof. $\int (bx-1)e^{-x}\mathrm{d}x = b - 1$; $\int (bx-1)^2 e^{-2x}\mathrm{d}x = b^2\tfrac{1}{4} - 2b\tfrac{1}{4} + \tfrac{1}{2}$. □

Corollary 2. *Any profile of the form $u = 1 - \mu''$, with μ the enclosed-mass fraction of* any *disc whose annulus mass μ' vanishes at the centre and at infinity, conserves mass at first order, since $\int_0^\infty \mu''\mathrm{d}x = [\mu']_0^\infty = 0$. The selection of $b = 1$ is the statement that the first-order deviation is the gradient of an annulus mass.*

Proposition 8 (exact split of the displaced mass)**.**

$$F(x) = x - m(x) = 2\mu'(x) - \int_0^x \mu''(t)^2 \,\mathrm{d}t. \tag{6}$$

The first term redistributes mass locally and vanishes far out; the second grows monotonically to $c = \frac{1}{4}$*. At* r_c*:* $0.736 - 0.216 = 0.520$*.*

Proof. $F' = 1 - u^2 = -2g - g^2 = 2\mu'' - \mu''^2$, and $F(0) = 0 = \mu'(0)$; integrate. □

5 A first-order reading: displacement and energy

This section is an interpretation, valid where $|g|$ *is small (beyond about* $1.5\,r_c$*, where* $|g| \le 0.14$*); it is not a derivation.* In an SIS, $\mathrm{d}M/\mathrm{d}R = v^2/G$, so displacing every shell outward by $\xi(R)$ changes the enclosed mass by $-(v^2/G)\,\xi$. The first-order part of F, $2\mu'$, therefore corresponds to

$$\xi(R) = 2r_c\,\mu'(x) = 2R\,e^{-R/r_c}. \tag{7}$$

A circular orbit in the SIS potential $v^2 \ln R$ has specific energy $v^2(\frac{1}{2} + \ln R)$, so moving it by ξ costs $v^2\xi/R$:

$$\varepsilon(R) = \frac{v^2\,\xi}{R} = 2v^2 e^{-R/r_c} = 2v^2\,\frac{\Sigma_d(R)}{\Sigma_d(0)}. \tag{8}$$

At first order, P18 is the SIS whose dark matter has received, at each radius, an energy per unit mass proportional to the local surface density of an exponential disc with $R_d = r_c$. The amplitude is not free: a profile $u = 1 - A\mu''$ is regular ($u(0) = 0$) only for $A = 1$, which gives $u'(0) = 2$, $r_c = R_d$, and $\varepsilon(0) = 2v^2$. This agrees in order with the requirement found in the companion note *Halting and settling*, that the most bound matter of the SIS be lifted by an energy of order v^2. The total, $\int \varepsilon\,\mathrm{d}M_{\rm SIS} = 2v^2 \cdot v^2 r_c/G$, is $6 \times 10^{56}\,\mathrm{erg}$ for the median dwarf ($v = 73\,\mathrm{km\,s^{-1}}$, $r_c = 2.4\,\mathrm{kpc}$): the supernova energy of about $6 \times 10^7\,\mathrm{M}_\odot$ of stars at full coupling. Two cautions apply. At the centre the displacement is non-linear ($\xi/R \to 2$). And a collisionless response is not a local displacement: each orbit averages over its radial range, so a local rule requires nearly isotropic orbits.

6 The anisotropy P18 requires

Proposition 9. *Let the radial velocity dispersion be constant at the SIS value,* $\sigma_r^2 = v^2/2$*. Then P18 is an equilibrium of the spherical Jeans equation,* $\mathrm{d}(\rho\sigma_r^2)/\mathrm{d}R + 2\beta\rho\sigma_r^2/R = -\rho GM/R^2$*, if and only if*

$$\beta(x) = 1 - \frac{x\,u'}{u} - \frac{m}{x}. \tag{9}$$

$\beta \simeq \frac{3}{4}x$ *for* $x \to 0$ *and* $\beta \simeq -c/x$ *for* $x \to \infty$*.*

Proof. With σ_r constant, $\sigma_r^2(\mathrm{d}\ln\rho/\mathrm{d}\ln R + 2\beta) = -V^2$. From Eq. (1), $\mathrm{d}\ln\rho/\mathrm{d}\ln R = 2xu'/u - 2$, and $V^2 = v^2 m/x$; inserting $\sigma_r^2 = v^2/2$ gives Eq. (9). Near the centre $u = 2x - \frac{3}{2}x^2 + \ldots$, so $xu'/u = 1 - \frac{3}{4}x + \ldots$, and $m/x = O(x^2)$. Far out $u' \to 0$ exponentially and $m/x = 1 + c/x + \ldots$. □

Table 2: The anisotropy of Eq. (9). Maximum 0.161 at $x = 2.47$; minimum -0.021 at $x = 10.3$.

x	0.1	0.3	0.5	1	1.5	2	3	4	6	10
β	0.062	0.129	0.152	0.152	0.150	0.157	0.155	0.114	0.022	−0.021

At fixed temperature, P18 is the SIS plus a modest radial bias in the zone where it has been rearranged, with isotropic orbits at the centre (as the distribution functions of *Halting and settling* also require). The Jeans equation can support many profiles with a suitable β, so Eq. (9) is a target for a mechanism, not an explanation.

Remark (general relativity). The weak-field limit of the Einstein equations for a static sphere with radial and tangential pressures, $p_r = \rho\sigma_r^2$ and $p_t = \rho\sigma_t^2$, is exactly this Jeans equation. At $v = 73\,\mathrm{km\,s^{-1}}$, $v^2/c^2 \simeq 6 \times 10^{-8}$, so relativistic corrections to the core are negligible. Spherical symmetry forces $p_\theta = p_\varphi$; a disc requires an axisymmetric treatment, whose Newtonian form is the axisymmetric Jeans equations with σ_R, σ_φ, σ_z and a mean rotation.

7 The disc in the collapse of the paper

The dark matter stays spherical. Each shell of the seed receives by turnaround the specific angular momentum $j = 2\sqrt{2}\lambda GM/v$ (mean over the mass inside it: $\sqrt{2}\lambda GM/v$, paper Sect. 6.5). A collisionless shell contracts by a factor of only 2 between turnaround and halting, to $R \simeq GM/v^2$, so its rotation is at most $j/R \simeq 2\sqrt{2}\lambda v$, against $\sigma = v/\sqrt{2}$ in the SIS: $v_\mathrm{rot}/\sigma \lesssim 4\lambda = 0.10$–$0.20$ for $\lambda = 0.025$–0.05. The flattening, of order $(v_\mathrm{rot}/\sigma)^2$, is a few per cent at most.

The gas forms a disc near r_c. Gas radiates and contracts until it is supported by rotation, at its centrifugal radius. With the paper's disc relation, $R_d/r_c = f_j\lambda y/\sqrt{2}$ (Mo–Mao–White), and the factor 1.06–1.4 found for the disc of the seed (companion note *The disc*), the core ratios y of the paper's Table 1 give:

λ	$y(z{=}0.5)$	λy	R_d/r_c ($f_j = 1$)
0.025	39	0.98	0.69–0.97
0.035	34	1.19	0.84–1.18
0.05	28	1.40	0.99–1.39

Since y falls as λ rises, λy varies only as $\lambda^{0.15\text{–}0.5}$, and R_d/r_c stays of order unity. Both R_d and r_c grow as t, so the ratio holds during growth. The observed stellar discs have a median $R_d/r_c = 0.67$ with 0.30 dex scatter.

The mass budget. The cosmic baryon fraction is $\Omega_b/\Omega_m = 0.157$. In units of $v^2 r_c/G$ (the SIS mass inside r_c is 1):

all baryons inside the front, $0.157\,y$ ($y = 34$)	5.35
baryons the dwarfs keep, 1.7% of the mass inside the front (paper)	0.58
mass P18 removes from inside r_c, $F(1)$	0.52

A thin exponential disc has $V_\mathrm{max}^2 = 0.387\,GM_d/R_d$ at $2.15\,R_d$. With all baryons and $R_d = r_c$ the disc alone would reach $1.44\,v$, a baryon-dominated, peaked curve; with the retained baryons, $0.47\,v$. About 90% of the gas that passes through the disc must therefore leave. Lifting it from r_c to the front costs $v^2 \ln y = 3.5\,v^2$ per unit mass, in total $16.8\,v^2(v^2 r_c/G)$, eight times the $2\,v^2(v^2 r_c/G)$ of Eq. (8). That the kept disc is comparable to the displaced mass is noted, not claimed: the model's value depends on y, which is uncertain by a factor of a few.

Two time scales. The halted shells of the paper lie at the front, $R_b = \xi_b v t$ with $\xi_b = 1/(1 + 3\pi/2) = 0.175$; their gas continues inward to $\sim r_c = R_b/y$. The inflow therefore runs on the growth time t, while an orbit at r_c takes $2\pi r_c/v = 2\pi\xi_b t/y = 0.03\,t$: the dark matter in the core responds *adiabatically* to the growing disc and contracts. Star-formation bursts last tens of Myr, against an orbital period at r_c of about 200 Myr for the median dwarf: the outflows act *impulsively*.

The hypothesis tested. (1) Dark matter and 16% gas collapse; the dark matter halts near the front, spherical. (2) The gas settles into a disc with $R_d \approx r_c$, which supplies the pattern $\Sigma_d \propto e^{-R/r_c}$ and, through Eq. (4), the critically damped form. (3) Stars ignite in the disc, and about 90% of the gas is expelled in cycles. (4) The spherical dark matter answers this forcing with $\varepsilon \propto \Sigma_d$. (5) What remains is P18 total matter. Steps (1)–(3) follow from the paper's numbers; step (4) is tested below.

8 Numerical experiments

8.1 Bursts in the self-similar shells

The earlier note *u_oscillator* (subfolder `u_oscillator` of this package) adds to the paper's one-dimensional self-similar shells (2000 shells, $\lambda = 0.035$, $j = \sqrt{2}\lambda R_{\rm ta}$ at turnaround, units $G = v = 1$) a prescribed central baryonic Plummer sphere of mass $f_b\,\xi_f\,t$ ($\xi_f = 0.2$), whose radius jumps between $0.002\,t$ and $a_e\,t$ in cycles of length $P\,t$ (code: `code/shells/fbshells.py`). The front lies at $\xi = R/vt \simeq 0.21$, the model's core at $\xi \simeq 0.006$. Table 3 summarises the nine runs.

Table 3: Bursts in the self-similar shells (from the earlier note *u_oscillator*). Slope: dark-matter density at $\xi = 0.002$–0.006. F: displaced dark matter against the run without baryons.

run	f_b	a_e	P	slope	peak of F (at ξ)	$F < 0$ beyond ξ
reference	0	–	–	-2.46	–	–
fb05, fb10, fb20	0.05–0.20	0.05	0.05	-1.28, -0.77, -1.26	0.04–0.09 (0.20–0.22)	–
fb05fast	0.05	0.05	0.01	-1.02	0.051 (0.20)	–
fb10far	0.10	0.20	0.05	-1.44	0.072 (0.22)	–
fb05ae02, ae01	0.05	0.02, 0.01	0.05	-1.17, -1.80	0.035, 0.020 (0.20–0.22)	0.61, 0.48
fb02ae01	0.02	0.01	0.05	-2.27	0.006 (0.09)[a]	0.28

[a] Only 3–5 times the noise of F.

The bursts flatten the cusp and rearrange the dark matter with a P18-like shape (a single peak of F and, when the gas is confined, a single zone of accumulation), but always on the scale of the front, about $35\,r_c$, never at r_c. On nearly radial orbits the displaced matter is deposited near its apocentres, where the infall turns it round.

8.2 Disc bursts in an isotropic SIS

Model (`code/discburst.py`). Dark matter: 2×10^5 particles in three dimensions, initially an isotropic SIS with $v = 1$ ($\rho = 1/4\pi r^2$, Maxwellian velocities with $\sigma = 1/\sqrt{2}$), truncated at $R_{\rm out} = 100$, with a spherical force from the enclosed dark mass (Plummer softening 0.02), leapfrog with $\Delta t = 0.004$. Gas: an exponential disc of scale length $R_d = 1$, entering through its spherically enclosed mass $M_d(t)\,\mu(r)$. Schedule: relaxation for 10 time units; the disc is built up to the retained mass $M_{\rm lo} = 0.5$ over $T_{\rm fill} = 20$ (three orbital periods at R_d); then bursts, each a linear filling by ΔM over $T_{\rm fill}$ (adiabatic) followed by an instantaneous drop back to $M_{\rm lo}$ (impulsive); finally $M_{\rm lo}$ is kept. All runs last 200 time units and are compared, over 15 snapshots in the last 18 units, with the run without disc. Units $G = v = R_d = 1$, so masses are in $v^2 R_d/G$.

Table 4: Disc bursts in an isotropic SIS (`code/analyse_out.txt`). Expelled: total gas mass removed by the bursts. Slope: dark-matter density at $R = 0.05$–$0.2\,R_d$ (reference -2.04). $\rho/\rho_{\rm ref}$: dark matter against the run without disc. r_c: P18 fitted to dark matter plus retained disc.

run	bursts $\times\,\Delta M$	expelled	slope	$\rho/\rho_{\rm ref}$ at $R = 0.3, 1, 3\,R_d$	β at $1, 3\,R_d$	r_c/R_d
adiab (disc grown, kept)	–	0	-2.06	1.12, 1.12, 1.06	0.01, 0.01	0.05
cyc03	7×0.3	2.1	-1.90	1.12, 1.12, 1.06	0.01, 0.00	0.05
cyc06	7×0.6	4.2	-1.81	1.09, 1.08, 1.05	0.00, 0.01	0.05
big3	3×1.5	4.5	-1.91	1.09, 1.05, 1.00	0.05, 0.04	0.05
strong7	7×3.0	21	-1.97	0.96, 0.66, 0.62	0.17, 0.25	0.04

Results (Table 4, Fig. 2).

1. *Within the budget nothing much happens.* The retained disc, grown slowly, contracts the dark matter by about 10% between 0.3 and $2\,R_d$. Bursts that expel up to 4.5 (the budget of Sect. 7 is 4.7) only partly undo this contraction; the inner slope changes from -2.04 to

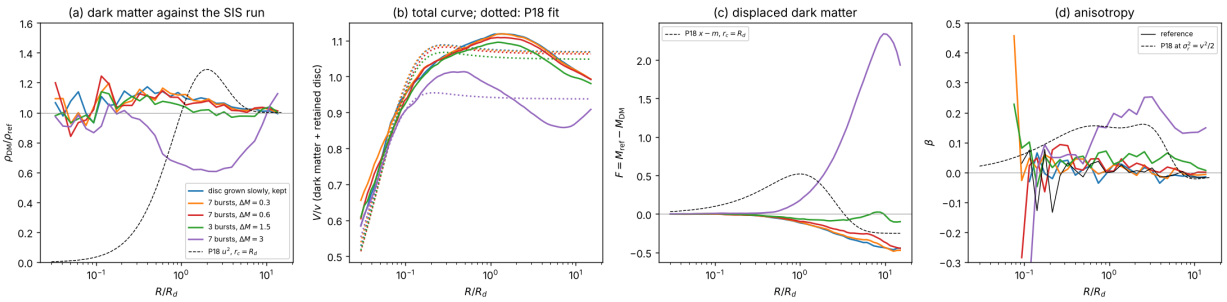


Figure 2: Disc bursts in an isotropic SIS. (a) Dark-matter density against the run without disc, with P18's u^2 for $r_c = R_d$ (dashed). (b) Circular speed of dark matter plus retained disc; dotted: P18 fits. (c) Displaced dark matter, with P18's $x - m$ (dashed). (d) Anisotropy, with the reference run and Eq. (9) (dashed). Script: `code/fig_runs.py`.

between -1.8 and -1.9. Fewer and larger bursts (big3) do somewhat better than many small ones with the same total, as expected for a heating of second order.

2. *Strong forcing expands the halo but keeps the cusp.* With 21 units expelled (strong7), far beyond the budget, the dark matter is reduced by 35–40% between 1 and 5 R_d and moved beyond 10 R_d; the orbits there become radially biased ($\beta \simeq 0.2$, the size Eq. (9) asks for). But the inner slope stays at -1.97: no core, no crossing and no overshoot on the scale R_d.

3. *No P18 on the scale of the disc.* Fitted to dark matter plus retained disc, P18 returns $r_c \simeq 0.05\,R_d$ in every run, i.e. the softened cusp, not a core of size R_d.

8.3 Why the centre is shielded

Proposition 10. *Let the dark matter be an SIS, $M_{\rm DM}(<r) = v^2 r/G$, and let an exponential disc of mass δM and scale length R_d be added or removed, acting through its spherically enclosed mass. Then the relative change of the enclosed mass, and of the gravitational force, is*

$$\frac{\delta M\,\mu(r/R_d)}{M_{\rm DM}(<r)} = \frac{G\,\delta M}{v^2 R_d}\,\frac{\mu(x)}{x} \simeq \frac{G\,\delta M}{v^2 R_d}\,\frac{x}{2} \qquad (x = r/R_d \to 0), \tag{10}$$

and the change of potential across an orbit of size r is, relative to v^2, of the same order. If instead $M_{\rm DM} \propto r^2$ (an inner density $\propto r^{-1}$), the ratio tends to a constant.

Proof. $\mu(x) = x^2/2 + O(x^3)$ from Lemma 1. The force ratio equals the enclosed-mass ratio in spherical symmetry. The added force near the centre is the constant $G\,\delta M/(2R_d^2)$, so the potential changes across an orbit of size r by $G\,\delta M\,r/(2R_d^2)$, which relative to v^2 is the same ratio. For $M_{\rm DM} = kr^2$ the ratio tends to $G\,\delta M/(2kR_d^2)$. □

The heating by a cycle of slow filling and sudden emptying is of second order in the potential change, so it vanishes even faster towards the centre. A disc-shaped forcing therefore cannot lift the most bound dark matter of an SIS by $\varepsilon(0) = 2v^2$, as Eq. (8) requires, whatever its amplitude; the runs confirm this. Feedback cores in simulations start from cusps with $\rho \propto r^{-1}$, for which the ratio stays finite. The collapse of the paper, with the shells left to themselves, ends in an SIS-like cusp (paper, Appendix H), which is much harder to erase.

9 Conclusions and open problems

Established. The factor u is the regular, critically damped solution with $b = 1$; it equals $1 - \mu''$ for an exponential disc with $R_d = r_c$; the double root is the composition of the exponential surface density with the annulus factor; $b = 1$ is singled out by mass conservation at first order; the excess

$c = \frac{1}{4}$ is second order; and at the SIS temperature P18 requires a radial bias $\beta \simeq 0.15$ between 0.5 and $3\,r_c$ with isotropic orbits at the centre. In the paper's collapse a gas disc with $R_d \approx r_c$ forms naturally, and most of its gas must be expelled.

Excluded (within the approximations stated). The dark matter cannot be given the P18 form by the changing potential of central gas bursts alone: in the self-similar shells the rearrangement sits at the front; in an isotropic SIS a disc-shaped forcing cannot reach the centre (Proposition 10).

Open, for a future project:

1. *A core that never becomes a cusp.* If the disc grows while the halting shells pass through the centre, the SIS cusp need not form at all. This is assumption (A4) of the paper with a disc in it, and it fits the paper's picture of a core that grows with the galaxy. It requires the shell code with a disc grown during the collapse.

2. *Torques instead of mass changes.* A rotating, non-axisymmetric disc (bar, spiral arms) exchanges angular momentum with dark-matter orbits through resonances; such torques are known to flatten cusps and isotropise orbits (Weinberg & Katz 2002, ApJ 580, 627). A first test: test particles in an SIS plus a rotating bar of length $\sim R_d$.

3. *A local coupling rule.* Eq. (8) asks for an energy per unit dark mass proportional to Σ_d. A naive feedback estimate gives energy per unit area $\propto \Sigma$ but, divided by the dark column of an SIS ($\propto 1/R$), an energy per unit mass $\propto R\Sigma$. Which physics, if any, gives Σ itself is open.

4. *The exact seed disc.* The disc of the seed is only nearly exponential (twice the exponential mass inside $0.5\,R_d$, a sharp edge at $4.3\,R_d$). Its $1 - \mu''$ should be compared with u.

References

Background: Blumenthal et al. (1986, ApJ 301, 27) for adiabatic contraction; Navarro, Eke & Frenk (1996, MNRAS 283, L72) and Pontzen & Governato (2012, MNRAS 421, 3464) for cores from mass loss and potential fluctuations; Mo, Mao & White (1998, MNRAS 295, 319) for disc sizes; Binney & Tremaine (2008, *Galactic Dynamics*) for the Jeans equations.

Files in this package

`code/identities.py`	checks of every formula and number of Sects. 2–7; output `identities_out.txt`
`code/fig_disc.py`	Fig. 1
`code/discburst.py`	the disc-burst model of Sect. 8.2
`code/runall.py`	runs it (`python3 runall.py [names]`, two at a time); log `runs.log`
`code/analyse.py`	analysis (output `analyse_out.txt`)
`code/fig_runs.py`	Fig. 2
`code/shells/fbshells.py`	the self-similar shell code with central bursts of Sect. 8.1; its nine runs, their analysis scripts, run files and figures are in the subfolder `u_oscillator`

The subfolder `u_oscillator` holds the earlier, more detailed note on the differential equation and the burst experiment in the self-similar shells (Sects. 2 and 8.1 here), with its code and run files. The run files of Sect. 8.2 (about 13 MB each) are not included; the runs take about 10 minutes each and reproduce the numbers exactly with the seed set in the code.